Engine Brief: Hotel & Assembly Building Types

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Adding hotel and assembly (place-of-worship) floor plans needs one shared capability layer plus two type configs on top. None of it exists today, and building the shared layer once also unlocks factories, schools and banks later.

Shared foundation (both types need these)

Five capabilities sit under both briefs. None exists today; building them once unlocks both types plus factories, schools and banks later.

#	Capability	What it does	Build note
A	Honest-output gate	Requires a valid, area-checked plan to exist before anything renders. No plan → no 3D picture; the pre-screen returns "not supported yet" or "couldn't fit," never a false "pass."	Do first — small, independent, and it stops misleading results today.
B	Non-residential room-type library	A typed catalogue so unknown rooms stop defaulting to *private*. Each entry carries category (habitable / wet / service / circulation / assembly), wet-room flag, min area and proportions, and adjacency rules.	Cheap config once the structure exists. Add the hotel and assembly vocabularies per brief.
C	Occupancy engine	Takes a capacity (guests, seats) and computes exits, exit widths, stair count and restroom fixtures by sex.	Constants (occupant-load, exit-width-per-person, fixtures-per-N) must be read from the Nigerian National Building Code / Lagos code — a lookup table, not hardcoded guesses.

#	Capability	What it does	Build note
D	Structural-system selector	Replaces the single 6 m bay rule with a choice: *framed* (≤6 m bays) for cellular layouts, or *long-span* (portal frame / truss, ~12–30 m) for halls.	A room flagged "assembly" or "wide-span" routes to long-span instead of failing the span check.
E	Zoning / setback categories	Adds **hospitality** and **place-of-worship / assembly** beyond residential and Lagos-commercial.	Each needs a cited setback source; where Lagos has no assembly category, document the fallback rather than silently using commercial.

Brief 4 — Hotel / lodging

Why it fails now: every guest-service room defaults to private, the restaurant wants a span over 6 m, and there is no guest-count logic for egress.

Room-type additions: reception, lobby, lounge/bar, restaurant (assembly / long-span flag), back-of-house kitchen (wet), guest room, suite, housekeeping store (service), lift core, admin office, staff facilities, laundry (wet).

Type-specific logic to add:

1. Double-loaded corridor generator — place a repeating room module along a central circulation spine, not each room as a free object.
2. Typical-floor repetition — design one guest floor, stack it N times with the core aligned, and generate a distinct ground floor underneath (extends the flats stacking already proven).
3. Vertical transport — a lift core plus a code-required number of fire-escape stairs, sized by occupancy (capability C).
4. Public / back-of-house separation — an adjacency rule so guests and service do not share circulation.

Acceptance criteria:

- [ ] Generates a repeatable guest floor with rooms off a central corridor, core aligned floor-to-floor.
- [ ] Restaurant renders as one clear span with no false span warning.
- [ ] Exit and stair counts change correctly when guest count changes.

- [ ] Refuses honestly if the program cannot fit, with no render.

Reference prompt it should eventually pass: the original boutique-hotel prompt (4-storey, 30 × 40 plot, 15 rooms/floor, restaurant, lift + fire stair), unchanged. That prompt *is* the acceptance test.

Brief 5 — Assembly / place-of-worship

The heavier build — a project of its own.

Why it fails now: no assembly setback, every auditorium room defaults to private, the 25 m hall blows the 6 m cap, and 1,000-seat exits and restrooms are not computed.

Room-type additions: auditorium / main hall (assembly + long-span), stage/altar, choir area, instrument store, overflow / children's section, entrance foyer, vestry, office, ushers' room, bookshop, crèche / nursing room (wet), congregation restrooms (wet, fixture-count driven).

Type-specific logic to add:

1. Seating-layout module — from a seat count, lay out banked seating with a central and two side aisles, code-width aisles, and a clear sightline to the stage. The core new capability and the hardest.
2. Capacity-driven egress — seat count drives minimum exit count, total exit width and exit distribution around the hall (capability C). Four exits is a computed minimum, not a fixed assumption.
3. Long-span roof — route the hall to the portal / truss system (capability D), sanity range ~15–30 m instead of the 6 m rule.
4. Gallery / mezzanine — an optional upper level over the foyer for offices, bookshop and restrooms, so support spaces do not eat the main-floor footprint.
5. Fixture sizing — restrooms sized to congregation via capability C, not a fixed room.

Acceptance criteria:

- [ ] Lays out ~1,000 seats with code-compliant aisles and a clear stage sightline.
- [ ] Hall generates a column-free (or minimal-column) long-span roof with no false span warning.
- [ ] Exit number / width and restroom fixtures scale with seat count.
- [ ] Uses an assembly setback with a documented source, not commercial-by-default.
- [ ] Support spaces fit, using a gallery when the ground floor is tight — or refuses honestly.

Reference prompt it should eventually pass: the original 1,000-seat church prompt, unchanged.

Build order

Cheapest and least misleading first; the genuinely new work last.

1. **Capability A — honest gate.** Small, independent, stops false passes and word-only renders today.
2. **Capabilities B + E — room types and zoning.** Cheap config once A exists; no new models.
3. **Hotel.** The smaller leap — reuses existing flats stacking; only needs the corridor generator and the occupancy engine (C).
4. **Assembly / church.** Last — the seating module and long-span roof are genuinely new. The flagship, built once the rest is proven.